

# Tirtharaj Barma

📞 +91 8116-841528 — ✉ tirtharajbarma3@gmail.com — 🌐 <https://www.linkedin.com/in/tirtharajbarma>  
📄 <https://github.com/TirtharajBarma>

**About me** — I am a full-stack developer with experience in building scalable web applications using React, Node.js, MongoDB, and Express. Skilled in creating user-centric UIs, integrating REST APIs, and deploying high-performance solutions using AWS and DevOps best practices.

## Skills

|                   |                                                                             |
|-------------------|-----------------------------------------------------------------------------|
| Languages         | JavaScript(ES6+), TypeScript, C++, Java, SQL                                |
| Web Frameworks    | React, Redux, Node.js, Express, Tailwind                                    |
| Database & tools  | Mongodb, MySQL, Mongoose                                                    |
| Development tools | Git, GitHub, Docker, AWS                                                    |
| CourseWork        | Object-Oriented Programming, Networking, DBMS, Data Structure and Algorithm |

## Experience

**AWS Cloud Club LPU** November 2023 – Current

*Core Team Member (Web development)*

- I am selected as a core team member who will lead initiatives promoting AWS Cloud awareness and skill development in students.
- Successfully organized and executed two large-scale campus events (technical workshops and hands-on labs) resulting in 30% more students participating and engaging.
- Peer mentoring led to collaborative learning, breaking barriers around simplified learning about cloud computing concepts and AWS solutions, translating to a 50% improvement in student comprehension and confidence.

## Projects

**Real-Time Chat Application — MERN stack** January 2025 – Current

- Developed a feature-rich real-time chat application with seamless, lag-free communication using **WebSocket**, enabling instant message delivery and improving user satisfaction by 70%.
- Integrated **JWT-based** authentication to enable the user to securely log in, keeping data safe and user experience improved
- Integrated typing indicators, **online/offline status**, and real-time updates to provide an interesting and interactive user experience.

**Crypto Tracker Website — React, Responsive Web Design** September 2024 – November 2024

- Built responsive and visually stunning cryptocurrency tracker, aimed at delivering live price updates and market data for mainstream cryptocurrencies, ensuring 95% satisfaction from users.
- Integrated **Google Charts** to create interactive visualizations that provide users a genuine overview of trends concerning cryptocurrency prices across different timeframes.
- Increased user engagement by 65% through the addition of detailed cryptocurrency descriptions, **historical data**, and advanced trend analysis tools driving user retention.

**Book-Store WebApp — MERN Stack, Firebase** April 2024 – June 2024

- An intuitive Admin Dashboard designed with fluid CRUD functionality enables the easy addition, edit, and delete of book records for 70% operational efficiency.
- Created synchronous real-time data synchronization that reflects newly-added books on front-end website in real-time, improving trust among users while reducing delays by 80%.
- **Google Firebase Authentication** for secure user authentication and robust session management, which in turn caused a 90% reduction in incidents of unauthorized access.

## Education

**Lovely Professional University** Sep 2022 - July 2026

*B.Tech. in Computer Science and Engineering*

*CGPA 7.76*

*Minors: Web Programming and Devops*

### Certifications

- Cloud Computing - NPTEL (2024)
- Server Side Scripting with Node.js - Coursera (2024)
- Data Structures and Algorithms Using C++ - Udemy (2023)